

Lab 3 – Architecture Selection & Written Justification

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Architecture Selection: We chose Microservices Architecture for the Municipal Infrastructure Damage Reporting App.

Architectural Choice: Microservices is suitable because the system contains distinct functions such as report submission, automatic damage classification and routing, GIS/location handling, notifications, and report storage. These functions can be separated into independently managed components communicating through defined interfaces.

Reason 1 – Independent Scaling and Performance: The classification and routing service may receive many reports and can be scaled independently from the citizen application or notification service. This supports the requirement that tickets are classified and routed to the correct municipal department within 5 minutes and avoids making the whole application scale as one unit.

Reason 2 – Fault Isolation and Modularity: A failure in a secondary service, such as notifications, should not prevent a citizen from creating and tracking a damage report. Separating services also makes it easier to modify classification, GIS integration, or notification mechanisms without redesigning the entire application.

Security Advantage: Each service can enforce authenticated and authorized API access, while communication between services can use TLS 1.2+. Stored photos and GPS/location data can be protected using AES-256, matching the system's security requirement.

Performance Benefit: Services can process work independently and communicate through focused APIs/events. This reduces unnecessary processing in the citizen-facing component and allows high-load services such as classification/routing to be scaled separately.